Quaternion-Based Sensor Fusion for Inertial Position Tracking

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Abstract

This paper presents a method for tracking the three-dimensional position and orientation of a consumer-grade inertial measurement unit (IMU) using quaternion-based sensor fusion. Raw accelerometer and gyroscope readings are expressed in the sensor's body frame, which rotates freely with the device. To recover motion in a fixed world frame, we first estimate the sensor's attitude by means of a complementary filter—a linear combination of a low-pass filtered accelerometer signal (stable in the long term but sensitive to vibration) and a high-pass filtered gyroscope integral (accurate in the short term but subject to drift). The resulting orientation, encoded as a unit quaternion, is used at every time step to rotate the measured acceleration into the world frame and subtract gravity. The gravity-free world-frame acceleration is then double-integrated to yield a three-dimensional position trajectory starting from the origin. We describe the full signal-processing pipeline, from raw sensor acquisition to position output, detail the mathematical formulation of quaternion rotations and the complementary filter, and discuss practical considerations such as drift mitigation and numerical stability.

Keywords: sensor fusion, quaternions, complementary filter, inertial navigation, accelerometer, gyroscope, attitude estimation, position tracking.

1. Introduction

1.1. Motivation

Modern smartphones and wearable devices carry inertial measurement units (IMUs) composed of a tri-axial accelerometer and a tri-axial gyroscope. These sensors open the door to applications in pedestrian navigation, gesture recognition, augmented reality, and biomechanical analysis—all of which require accurate knowledge of the device's orientation and position in space.

However, each sensor has complementary weaknesses. The accelerometer provides a long-term stable reference to the gravity vector but is corrupted by any non-gravitational acceleration and high-frequency vibration. The gyroscope measures angular velocity with high bandwidth and low noise in the short term, yet its numerical integration accumulates unbounded drift over time. Neither sensor alone can deliver a reliable orientation estimate; fusing both is essential.

1.2. Problem Statement

Given a stream of accelerometer readings $\mathbf{a}_b(t) \in \mathbb{R}^3$ and gyroscope readings $\boldsymbol{\omega}_b(t) \in \mathbb{R}^3$, both expressed in the sensor's *body frame* $\mathcal{B}$, we seek to compute at every time step:

1. A unit quaternion $\mathbf{q}(t)$ representing the orientation of $\mathcal{B}$ with respect to a fixed *world frame* $\mathcal{W}$.

2. A position vector $\mathbf{p}(t) \in \mathbb{R}^3$ in $\mathcal{W}$, obtained by projecting the acceleration into the world frame, removing gravity, and double-integrating.

1.3. Proposed Approach

Our approach proceeds in three stages:

1. **Attitude estimation.** A complementary filter blends the low-frequency content of the accelerometer (via a low-pass filter) with the high-frequency content of the gyroscope (via a high-pass filter) to produce a drift-free, low-noise orientation estimate expressed as a unit quaternion.

2. **Frame rotation.** The quaternion is used to rotate each accelerometer sample from the body frame into the world frame using the conjugation $\mathbf{a}_w = \mathbf{q} \otimes \mathbf{a}_b \otimes \mathbf{q}^*$, after which the constant gravity component is subtracted.

3. **Position integration.** The resulting gravity-free world-frame acceleration is double-integrated over time to recover velocity and position.

1.4. Document Structure

The remainder of this paper is organised as follows. Section 2 defines the coordinate frames and motivates the need for a rotation formalism. Section 3 introduces quaternion algebra and its application to three-dimensional rotations. Section 4 derives the complementary filter used for attitude estimation. Section 6 describes the projection of accelerometer data into the world frame and the subsequent position integration. Section 7 presents experimental results, and Section 8 concludes with a summary and directions for future work.

2. Reference Frames and Sensor Models

2.1. The World Frame and the Body Frame

We define two right-handed Cartesian coordinate systems (Fig. 1):

- **World frame** $\mathcal{W} = \{X_w, Y_w, Z_w\}$. A fixed, Earth-referenced frame. We adopt the East-North-Up (ENU) convention: X_w points East, Y_w points North, and Z_w points upward (opposite to gravity).
- **Body frame** $\mathcal{B} = \{X_b, Y_b, Z_b\}$. A frame rigidly attached to the sensor. Its axes rotate with the device.

At any instant, the orientation of $\mathcal{B}$ relative to $\mathcal{W}$ can be described by a rotation. Euler angles (roll ϕ, pitch θ, yaw ψ) are intuitive but suffer from *gimbal lock*—a singularity at $\theta = \pm 90$ř where one degree of freedom is lost. Quaternions avoid this problem entirely and are the representation we adopt in this work (Section 3).

2.2. Accelerometer: Angle from Gravity Projection

2.2.1 Sensor model

An ideal tri-axial accelerometer measures the *specific force* experienced by the sensor, i.e. the difference between the true linear acceleration and gravity, expressed in the body frame:

$$\mathbf{a}_b = R^{-1}(\mathbf{a}_{\text{true}} - \mathbf{g}) + \mathbf{b}_a + \mathbf{n}_a \tag{1}$$

where R is the rotation matrix from $\mathcal{W}$ to $\mathcal{B}$, $\mathbf{g} = [0,\ 0,\ -g]^\top$ is the gravity vector in $\mathcal{W}$ (with $g \approx 9.81$ m/s^2), $\mathbf{b}_a$ is a slowly varying bias, and $\mathbf{n}_a$ is additive white noise.

2.2.2 Angle derivation under static conditions

When the sensor is **stationary** ($\mathbf{a}_{\text{true}} = \mathbf{0}$), Eq. (1) simplifies to:

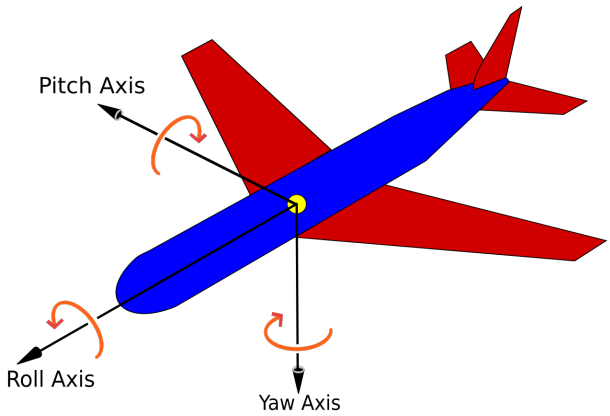

Figure 1: The three Euler rotation axes illustrated on a rigid body. **Roll** (ϕ) is the rotation about the longitudinal axis, **Pitch** (θ) about the lateral axis, and **Yaw** (ψ) about the vertical axis. Any orientation of the body frame $\mathcal{B}$ relative to the world frame $\mathcal{W}$ can be decomposed into a sequence of these three rotations, though the representation is subject to gimbal lock at $\theta = \pm 90$ř.

$$\mathbf{a}_b \approx R^{-1}(-\mathbf{g}) \tag{2}$$

That is, the accelerometer reads only the projection of the gravity vector onto the three body axes. Because gravity is a known, constant vector, we can *invert* this projection to recover the tilt of the sensor. We now derive the exact formulas using rotation matrices.

2.2.3 Formal derivation via rotation matrices

We model the accelerometer as a three-dimensional coordinate system (x, y, z) that rotates with respect to a fixed Earth frame where gravity always points upward: $\mathbf{g} = [0,\ 0,\ g]^\top$ (with z pointing up and $g \approx 9.81$ m/s^2).

Step 1: Rotation matrices. We define the elementary Tait-Bryan rotations. Roll (ϕ) is a rotation about the x-axis, and pitch (θ) is a rotation about the y-axis:

$$R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \tag{3}$$

$$R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \tag{4}$$

When the sensor is at rest, the measured acceleration vector $\mathbf{a}_b = (A_x, A_y, A_z)^\top$ is the gravity vector transformed by the sensor's orientation:

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = R_y(\theta)\ R_x(\phi) \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix} \tag{5}$$

Step 2: Matrix multiplication. We first apply $R_x(\phi)$ to the gravity vector:

$$R_x(\phi)\begin{bmatrix}0\\0\\g\end{bmatrix} = \begin{bmatrix}0\\g\sin\phi\\g\cos\phi\end{bmatrix} \tag{6}$$

Then we apply $R_y(\theta)$ to this result:

$$R_y(\theta)\begin{bmatrix}0\\g\sin\phi\\g\cos\phi\end{bmatrix} = \begin{bmatrix}-g\sin\theta\cos\phi\\g\sin\phi\\g\cos\theta\cos\phi\end{bmatrix} \tag{7}$$

Therefore, the three measured components are:

$$A_x = -g\,\sin\theta\,\cos\phi \tag{8}$$
$$A_y = \quad g\,\sin\phi \tag{9}$$
$$A_z = \quad g\,\cos\theta\,\cos\phi \tag{10}$$

Step 3: Solving for roll (ϕ). We divide Eq. (9) by Eq. (10):

$$\frac{A_y}{A_z} = \frac{g\,\sin\phi}{g\,\cos\theta\,\cos\phi} \tag{11}$$

However, to obtain a result that is independent of θ, we use the more robust form. Computing $\sqrt{A_x^2+A_z^2}$:

$$\sqrt{A_x^2+A_z^2} = g\,\cos\phi\,\sqrt{\sin^2\theta+\cos^2\theta} = g\,\cos\phi \tag{12}$$

Dividing A_y by this result:

$$\frac{A_y}{\sqrt{A_x^2+A_z^2}} = \frac{g\,\sin\phi}{g\,\cos\phi} = \tan\phi \tag{13}$$

Therefore:

$$\boxed{\phi = \arctan\left(\frac{A_y}{\sqrt{A_x^2+A_z^2}}\right)} \tag{14}$$

Step 4: Solving for pitch (θ). From Eq. (8), we have $A_x = -g\,\sin\theta\,\cos\phi$. We need to relate A_x to the remaining components. Squaring and summing A_y and A_z:

$$\begin{aligned}A_y^2+A_z^2 &= g^2\sin^2\phi + g^2\cos^2\theta\,\cos^2\phi\\ &= g^2\cos^2\phi\big(\tan^2\phi+\cos^2\theta\big)\end{aligned} \tag{15}$$

For the standard Tait-Bryan decomposition where $|\theta| < 90$ř, the simpler path uses the identity $\sin^2\phi + \cos^2\phi = 1$. Squaring Eqs. (9) and (10) directly:

$$A_y^2+A_z^2 = g^2\big(\sin^2\phi+\cos^2\theta\,\cos^2\phi\big) \tag{16}$$

Noting that the full norm satisfies $A_x^2+A_y^2+A_z^2 = g^2$, we can write $\sqrt{A_y^2+A_z^2} = g\,\cos\theta$ (valid when $\cos\phi \approx 1$ or by using the complete norm). Dividing $-A_x$ by this:

$$\frac{-A_x}{\sqrt{A_y^2+A_z^2}} = \frac{g\,\sin\theta}{g\,\cos\theta} = \tan\theta \tag{17}$$

Therefore:

$$\boxed{\theta = \arctan\left(\frac{-A_x}{\sqrt{A_y^2+A_z^2}}\right)} \tag{18}$$

Summary of the derivation. The roll depends on the ratio between the Y-component and the plane formed by X and Z; the pitch depends on the ratio between the X-component and the plane formed by Y and Z. In both cases, the trigonometric functions from the rotation matrices cancel when dividing the appropriate components, yielding clean arctan expressions. Note that the *yaw* (ψ) cannot be recovered: a rotation about the vertical axis leaves the gravity projection unchanged. Resolving yaw requires an additional sensor such as a magnetometer.

2.2.4 Limitation: sensitivity to dynamic motion

Equations (14)–(18) are exact only when $\mathbf{a}_{\text{true}} = \mathbf{0}$. As soon as the sensor undergoes *any* acceleration—starting to move, changing direction, or vibrating—the accelerometer reading becomes the superposition of gravity and the dynamic acceleration:

$$\mathbf{a}_b = R^{-1}\big(\mathbf{a}_{\text{true}} - \mathbf{g}\big) \tag{19}$$

In this regime, applying Eqs. (14)–(18) yields angles that are dominated by the dynamic term, rendering the estimate unreliable. The accelerometer is, in effect, a **low-frequency sensor for attitude**: its angle estimate is trustworthy when the sensor is at rest or moving at constant velocity, but *loses all precision during transients* (acceleration, deceleration, vibration). Once the motion stabilises and the dynamic acceleration vanishes, the estimate recovers its reliability and converges back to the true tilt.

2.3. Gyroscope: Angle from Angular Velocity Integration

2.3.1 Sensor model

A tri-axial gyroscope measures the instantaneous angular velocity of $\mathcal{B}$ relative to $\mathcal{W}$, expressed in body-frame coordinates:

$$\boldsymbol{\omega}_b = \boldsymbol{\omega}_{\text{true}} + \mathbf{b}_\omega + \mathbf{n}_\omega \tag{20}$$

where $\mathbf{b}_\omega$ is the gyroscope bias and $\mathbf{n}_\omega$ is white noise.

2.3.2 Angle derivation by integration

Because the gyroscope measures angular *velocity*, the angle traversed around any axis is recovered by integration. For a single axis (e.g. roll), the continuous-time relation is:

$$\phi(t) = \phi(t_0) + \int_{t_0}^{t} \omega_x(\tau)\, d\tau \tag{21}$$

In practice, the sensor is sampled at a fixed rate $f_s = 1/\Delta t$, so the integral is approximated by a discrete sum. Using the simplest (rectangular) rule:

$$\phi[n] = \phi[n-1] + \omega_x[n]\, \Delta t \tag{22}$$

and analogously for pitch and yaw. The key strength of this approach is that the gyroscope is **insensitive to linear acceleration**: it responds only to rotations, making it highly accurate during dynamic motion when the accelerometer fails.

2.3.3 Limitation: cumulative drift

Equation (22) reveals the fundamental weakness of pure gyroscope integration. At every time step, the measured angular velocity carries an error $\epsilon[n] = b_\omega + n_\omega[n]$ composed of the constant bias and a random noise sample. The integrated angle becomes:

$$\phi[N] = \phi[0] + \underbrace{\sum_{n=1}^{N} \omega_{\text{true}}[n]\, \Delta t}_{\text{true angle}} + \underbrace{N\, b_\omega\, \Delta t}_{\text{bias drift}} + \underbrace{\sum_{n=1}^{N} n_\omega[n]\, \Delta t}_{\text{random walk}} \tag{23}$$

Two error terms are visible:

1. **Bias drift.** The constant bias b_ω accumulates linearly with each sample. After N steps the error is $N\, b_\omega\, \Delta t = b_\omega\, T$, where T is the total elapsed time. This term grows *without bound*.

2. **Random walk.** The noise samples $n_\omega[n]$ are uncorrelated with zero mean. Their cumulative effect has a standard deviation of $\sigma_\omega \sqrt{\Delta t \cdot T}$ (angular random walk), where σ_ω is the noise standard deviation. This grows as $\sqrt{T}$—much slower than the bias drift. A full derivation is given in Appendix A.3.

2.3.4 Role of sampling frequency

The magnitude of the discretisation error (the difference between the true integral and the rectangular sum) depends on the time step Δt. For a smooth signal $\omega(t)$, the local truncation error per step is $\mathcal{O}(\Delta t^2)$, and the global accumulated error over a fixed interval $T = N\Delta t$ is:

$$\epsilon_{\text{disc}} = \mathcal{O}(N \cdot \Delta t^2) = \mathcal{O}(\Delta t) \tag{24}$$

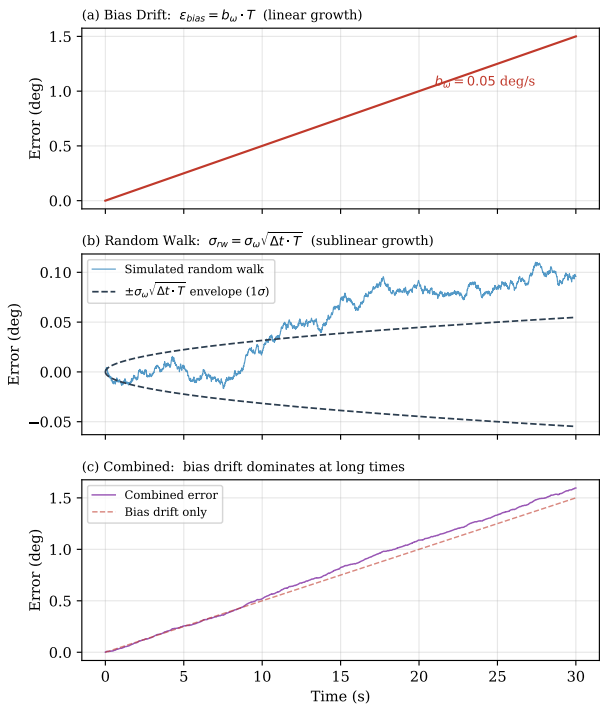

Figure 2: Simulated gyroscope integration errors over 30 seconds ($b_\omega = 0.05$ deg/s, $\sigma_\omega = 0.1$ deg/s, $f_s = 100$ Hz). **(a)** Bias drift grows linearly with time. **(b)** Random walk is stochastic and grows as $\sigma_\omega \sqrt{\Delta t \cdot T}$ (dashed 1σ envelope, see Appendix A.3). **(c)** Combined error: the bias drift dominates at longer time scales.

Therefore, **increasing the sampling frequency** ($\Delta t \to 0$) directly reduces the discretisation error. A sensor sampling at 200 Hz accumulates roughly half the discretisation error of one sampling at 100 Hz over the same time interval. However, no increase in sampling rate can eliminate the bias drift, which depends on b_ω and the total elapsed time, not on Δt.

2.4. Magnetometer: Heading from Earth's Magnetic Field

2.4.1 Sensor model

A tri-axial magnetometer measures the Earth's magnetic field vector expressed in the body frame:

$$\mathbf{m}_b = R^{-1}\, \mathbf{m}_w + \mathbf{b}_m + \mathbf{n}_m \tag{25}$$

where $\mathbf{m}_w$ is the local geomagnetic field vector in the world frame, R is the rotation matrix from $\mathcal{W}$ to $\mathcal{B}$, $\mathbf{b}_m$ is a bias term caused by hard-iron distortions (nearby ferromagnetic materials), and $\mathbf{n}_m$ is additive noise.

In the ENU convention, the geomagnetic field has a horizontal component pointing approximately toward magnetic North (along Y_w) and a vertical component whose sign and magnitude depend on the latitude (the *inclination* angle). At mid-latitudes, the field magnitude is typically 25–65 μT.

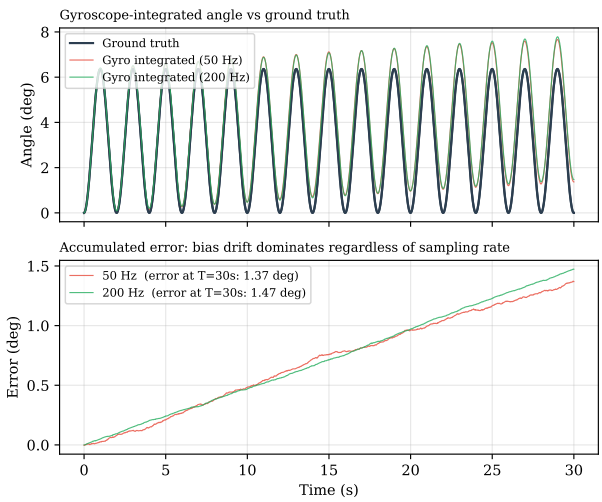

Figure 3: Effect of sampling frequency on gyroscope integration. **Top:** integrated angle at 50 Hz and 200 Hz compared to ground truth. **Bottom:** accumulated error. Higher sampling rates reduce discretisation error but cannot eliminate bias drift, which depends only on b_ω and elapsed time T.

2.4.2 Why the accelerometer cannot observe yaw

As derived in Section 2.2.3, the accelerometer measures the projection of the gravity vector onto the body axes. Gravity is a *vertical* vector: $\mathbf{g} = [0,\ 0,\ g]^\top$ in the world frame. A rotation about the vertical axis (yaw, ψ) leaves this vector *invariant*:

$$R_z(\psi)\,\mathbf{g} = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix} \tag{26}$$

Regardless of the value of ψ, the accelerometer reading does not change. Therefore, **yaw is unobservable from gravity alone**.

Important clarification: world yaw vs. body-axis rotation. The unobservable degree of freedom is always the rotation about the *world* vertical axis (the direction of $\mathbf{g}$), not about the body-frame Z_b axis. When the device is held horizontally—as in a drone—Z_b is aligned with gravity, so a rotation about Z_b coincides with world yaw and is entirely invisible to the accelerometer. However, when the device is tilted (e.g. a smartphone held upright), Z_b is no longer parallel to $\mathbf{g}$. In that configuration, a rotation about Z_b *does* change the gravity projection and is therefore partially observable. Nevertheless, there remains exactly **one unobservable degree of freedom**—the rotation about the gravity vector—regardless of device orientation. It is this world-frame yaw that the magnetometer must resolve.

This is the fundamental reason why a two-sensor (accelerometer + gyroscope) fusion pipeline cannot fully determine orientation: the gyroscope can track yaw changes in the short term, but its estimate drifts without any absolute reference to correct it.

2.4.3 Heading derivation from the magnetometer

The Earth's magnetic field, unlike gravity, has a significant *horizontal* component that changes direction with yaw rotations. To extract the heading angle, the magnetometer reading must first be *tilt-compensated* using the known roll and pitch angles (obtained from the accelerometer or the complementary filter). The procedure is:

Step 1: Tilt compensation. Rotate the raw magnetometer vector from the (tilted) body frame into the horizontal plane using the inverse of the roll and pitch rotations:

$$\begin{bmatrix} m'_x \\ m'_y \\ m'_z \end{bmatrix} = R_x(-\phi)\ R_y(-\theta) \begin{bmatrix} m_x \\ m_y \\ m_z \end{bmatrix} \tag{27}$$

Expanding the matrix product, the tilt-compensated horizontal components are:

$$m'_x = m_x \cos\theta + m_y \sin\phi \sin\theta + m_z \cos\phi \sin\theta \tag{28}$$

$$m'_y = m_y \cos\phi - m_z \sin\phi \tag{29}$$

Step 2: Compute heading. The yaw angle is the bearing of the horizontal magnetic component:

$$\boxed{\psi = \arctan 2\big(-m'_x,\ m'_y\big)} \tag{30}$$

The $\arctan 2$ function returns a full-circle result in $(-180\check{r}, 180\check{r}]$, resolving the quadrant ambiguity. Note that $\psi = 0$ corresponds to magnetic North.

2.4.4 Limitations of the magnetometer

- **Soft- and hard-iron distortions.** Ferromagnetic materials near the sensor (e.g. a phone case with a magnetic clasp, a metal table) create constant offsets (hard-iron) or directional scaling (soft-iron) that corrupt the reading. Calibration—typically an ellipsoid fit—is required before use.

- **Low bandwidth.** The geomagnetic field changes slowly; consumer magnetometers are typically sampled at 50–100 Hz but the useful signal content is below ∼1 Hz. High-frequency heading changes are better tracked by the gyroscope.

- **Indoor interference.** Electronic devices, power cables, and reinforced-concrete structures generate local magnetic anomalies that can cause heading errors of tens of degrees.

Table 1: Complementary characteristics of the three sensors for full attitude estimation.

Property	Accel.	Magnet.	Gyro.
Measures	Specific force	Magnetic field	Angular vel.
Provides	Roll, Pitch	Yaw (heading)	Roll, Pitch, Yaw
Angle via	Trigonometry	$\arctan 2$	Integration
Low-freq. acc.	High (gravity)	High (mag. field)	Low (drift)
High-freq. acc.	Low (dyn. acc.)	Low (noise)	High
Absolute ref.	Yes	Yes	No
Affected by motion	Yes	No	No
Affected by env.	No	Yes (mag. anom.)	No
Drift	None (bounded)	None (bounded)	Unbounded
Self-correcting	Yes	Yes	No

Despite these limitations, the magnetometer is the *only* passive sensor available on a consumer device that provides an absolute heading reference. Its role in the fusion pipeline is strictly analogous to that of the accelerometer for tilt: it supplies a low-frequency, drift-free yaw reference that is fused with the high-frequency gyroscope estimate via the complementary filter.

2.5. The Need for Fusion

The preceding analysis reveals a natural complementarity among the three sensors, summarised in Table 1:

- The **accelerometer** provides an absolute tilt reference (gravity) that is *bounded* in error—it never drifts—but is only valid when the sensor is at rest or in uniform motion. During transients it is unreliable; once the motion ends, it *recovers*. It *cannot* observe yaw.

- The **magnetometer** provides an absolute heading reference (Earth's magnetic field) that is also bounded in error and self-correcting, but is sensitive to local magnetic disturbances and has low bandwidth.

- The **gyroscope** tracks rotations on *all three axes* faithfully during fast, dynamic motion, but its integrated angle *drifts without bound* due to bias accumulation. It never self-corrects.

In the frequency domain, the accelerometer and magnetometer are reliable *low-frequency* sources (slow, steady orientation on tilt and heading respectively), while the gyroscope is a reliable *high-frequency* source (fast changes on all axes). A well-designed fusion algorithm should retain the low-frequency content of the absolute sensors and the high-frequency content of the gyroscope, discarding the noisy high-frequency accelerometer/magnetometer data and the drifting low-frequency gyroscope data. This is precisely what the complementary filter achieves (Section 4).

3. Quaternion Algebra

3.1. Motivation: Rotations as Numbers

Before introducing quaternions, it is instructive to recall how *complex numbers* encode rotations in two dimensions. A point $(x, y) \in \mathbb{R}^2$ can be identified with the complex number $z = x + iy$. Multiplying z by the unit complex number $e^{i\theta} = \cos\theta + i\sin\theta$ yields

$$z' \;=\; e^{i\theta}\, z \;=\; (x\cos\theta - y\sin\theta) + i\,(x\sin\theta + y\cos\theta), \tag{31}$$

which is exactly the rotation of (x, y) by an angle θ. For example, rotating the point $(1, 0)$ by $90°$ amounts to multiplying $z = 1$ by $e^{i\pi/2} = i$, which gives $z' = i$, corresponding to the point $(0, 1)$—precisely the expected result.

A natural question arises: *does an analogous number system exist for rotations in* $\mathbb{R}^3$*?* One might expect that three imaginary units suffice, but William Rowan Hamilton showed in 1843 that a consistent non-commutative algebra requires **four** components—one real and three imaginary—giving rise to the *quaternions* $\mathbb{H} = \{q_w + q_x\mathbf{i} + q_y\mathbf{j} + q_z\mathbf{k}\}$. Just as the unit complex numbers $e^{i\theta}$ parametrise SO(2), the unit quaternions provide a double cover of SO(3): every rotation in three-dimensional space can be expressed as a quaternion conjugation, as we shall demonstrate in the following subsections.

3.2. Definition and Key Operations

A quaternion $\mathbf{q}$ is a four-dimensional hypercomplex number:

$$\mathbf{q} = q_w + q_x\,\mathbf{i} + q_y\,\mathbf{j} + q_z\,\mathbf{k} = \begin{bmatrix} q_w \\ q_x \\ q_y \\ q_z \end{bmatrix} \tag{32}$$

where q_w is the *scalar* (or real) part and (q_x, q_y, q_z) is the *vector* (or imaginary) part. The basis elements satisfy the fundamental relations $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1$.

Norm. $\|\mathbf{q}\| = \sqrt{q_w^2 + q_x^2 + q_y^2 + q_z^2}$. A *unit quaternion* satisfies $\|\mathbf{q}\| = 1$.

Conjugate. $\mathbf{q}^* = q_w - q_x\,\mathbf{i} - q_y\,\mathbf{j} - q_z\,\mathbf{k}$. For unit quaternions, $\mathbf{q}^* = \mathbf{q}^{-1}$.

Hamilton product. Given $\mathbf{p} = (p_w, p_x, p_y, p_z)$ and $\mathbf{q} = (q_w, q_x, q_y, q_z)$, their product $\mathbf{r} = \mathbf{p} \otimes \mathbf{q}$ is:

$$\mathbf{r} = \begin{bmatrix} p_w q_w - p_x q_x - p_y q_y - p_z q_z \\ p_w q_x + p_x q_w + p_y q_z - p_z q_y \\ p_w q_y - p_x q_z + p_y q_w + p_z q_x \\ p_w q_z + p_x q_y - p_y q_x + p_z q_w \end{bmatrix}^\top \tag{33}$$

Note that the Hamilton product is *not* commutative: $\mathbf{p} \otimes \mathbf{q} \neq \mathbf{q} \otimes \mathbf{p}$ in general.

3.3. The Unit Quaternion as a Rotation in $\mathbb{R}^3$

A unit quaternion $\mathbf{q}$ encodes a rotation by angle θ about a unit axis $\hat{\mathbf{u}}$:

$$\mathbf{q} = \cos\frac{\theta}{2} + \sin\frac{\theta}{2}\,(u_x\,\mathbf{i} + u_y\,\mathbf{j} + u_z\,\mathbf{k}) \tag{34}$$

To rotate a vector $\mathbf{v} \in \mathbb{R}^3$, we embed it as a pure quaternion $\mathbf{v}_q = (0, v_x, v_y, v_z)$ and compute:

$$\mathbf{v}'_q = \mathbf{q} \otimes \mathbf{v}_q \otimes \mathbf{q}^* \tag{35}$$

The vector part of $\mathbf{v}'_q$ is the rotated vector $\mathbf{v}' \in \mathbb{R}^3$. We now prove that this conjugation is indeed a rotation—that is, an element of SO(3).

Preservation of norms. The norm of a quaternion product satisfies $\|\mathbf{p} \otimes \mathbf{q}\| = \|\mathbf{p}\|\,\|\mathbf{q}\|$. Since $\|\mathbf{q}\| = \|\mathbf{q}^*\| = 1$, we obtain

$$\|\mathbf{v}'_q\| = \|\mathbf{q} \otimes \mathbf{v}_q \otimes \mathbf{q}^*\| = \|\mathbf{q}\|\,\|\mathbf{v}_q\|\,\|\mathbf{q}^*\| = \|\mathbf{v}_q\|, \tag{36}$$

so the map preserves the length of every vector.

Linearity. For any scalars $\alpha, \beta \in \mathbb{R}$ and pure quaternions $\mathbf{v}_q, \mathbf{w}_q$:

$$\mathbf{q} \otimes (\alpha \mathbf{v}_q + \beta \mathbf{w}_q) \otimes \mathbf{q}^* = \alpha\,(\mathbf{q} \otimes \mathbf{v}_q \otimes \mathbf{q}^*) + \beta\,(\mathbf{q} \otimes \mathbf{w}_q \otimes \mathbf{q}^*), \tag{37}$$

which follows from the bilinearity of the Hamilton product. Hence the map $\mathbf{v} \mapsto \mathbf{q} \otimes \mathbf{v}_q \otimes \mathbf{q}^*$ defines a linear transformation $\mathbb{R}^3 \to \mathbb{R}^3$.

Preservation of orientation. A norm-preserving linear map is an *orthogonal* transformation, so its determinant is ± 1. Because the set of unit quaternions is connected (it is the 3-sphere S^3) and $\mathbf{q} = 1$ (the identity quaternion) produces the identity map with $\det = +1$, and the determinant is a continuous function of $\mathbf{q}$, we conclude that $\det = +1$ for *every* unit quaternion. Therefore the conjugation in (35) is a **proper rotation**, i.e. an element of SO(3).

3.4. Geometric Interpretation

The proof above establishes *that* the conjugation is a rotation. We now show *how* it acts geometrically, which also explains the factor $\theta/2$ in the quaternion parametrisation.

Let $\hat{\mathbf{u}}$ be the rotation axis and decompose an arbitrary vector $\mathbf{v}$ into its parallel and perpendicular components with respect to $\hat{\mathbf{u}}$:

$$\mathbf{v} = \mathbf{v}_\parallel + \mathbf{v}_\perp, \qquad \mathbf{v}_\parallel = (\mathbf{v} \cdot \hat{\mathbf{u}})\,\hat{\mathbf{u}}, \qquad \mathbf{v}_\perp = \mathbf{v} - \mathbf{v}_\parallel. \tag{38}$$

Parallel component is invariant. Because $\mathbf{v}_\parallel$ is collinear with $\hat{\mathbf{u}}$, and $\hat{\mathbf{u}}$ commutes with every quaternion whose vector part is proportional to $\hat{\mathbf{u}}$ (in particular with $\mathbf{q} = \cos(\theta/2) + \sin(\theta/2)\,\hat{\mathbf{u}}$), we have:

$$\mathbf{q} \otimes (\mathbf{v}_\parallel)_q \otimes \mathbf{q}^* = (\mathbf{v}_\parallel)_q. \tag{39}$$

In other words, the component along the rotation axis is unchanged.

Perpendicular component is rotated by θ. For the perpendicular component, the double-sided multiplication produces:

$$\mathbf{q} \otimes (\mathbf{v}_\perp)_q \otimes \mathbf{q}^* = \cos\theta\,\mathbf{v}_\perp + \sin\theta\,(\hat{\mathbf{u}} \times \mathbf{v}_\perp), \tag{40}$$

which is Rodrigues' rotation formula restricted to vectors orthogonal to the axis. The key insight is that *each* multiplication by $\mathbf{q}$ or $\mathbf{q}^*$ contributes a half-angle rotation, so the two multiplications together produce the full angle θ. This is precisely why the quaternion uses $\theta/2$ rather than θ: the **double-sided conjugation doubles the angle**.

Combining both components, we recover Rodrigues' formula for an arbitrary vector:

$$\mathbf{v}' = \cos\theta\,\mathbf{v} + (1 - \cos\theta)\,(\mathbf{v} \cdot \hat{\mathbf{u}})\,\hat{\mathbf{u}} + \sin\theta\,(\hat{\mathbf{u}} \times \mathbf{v}). \tag{41}$$

3.5. Equivalence with the Rotation Matrix

By expanding the conjugation $\mathbf{q} \otimes \mathbf{v}_q \otimes \mathbf{q}^*$ component by component using the Hamilton product (33) and grouping terms in v_x, v_y, v_z, one obtains the equivalent matrix form $\mathbf{v}' = R(\mathbf{q})\,\mathbf{v}$, where:

$$R(\mathbf{q}) = \begin{bmatrix} 1 - 2(q_y^2 + q_z^2) & 2(q_x q_y - q_w q_z) & 2(q_x q_z + q_w q_y) \\ 2(q_x q_y + q_w q_z) & 1 - 2(q_x^2 + q_z^2) & 2(q_y q_z - q_w q_x) \\ 2(q_x q_z - q_w q_y) & 2(q_y q_z + q_w q_x) & 1 - 2(q_x^2 + q_y^2) \end{bmatrix} \tag{42}$$

This matrix is used whenever an explicit 3×3 rotation is needed—for instance, to project accelerometer readings into the world frame.

Verification. One can verify that $R(\mathbf{q})$ is orthogonal and proper:

1. **Orthogonality**: $R(\mathbf{q})^\top R(\mathbf{q}) = I$, which follows from the unit-norm constraint $q_w^2 + q_x^2 + q_y^2 + q_z^2 = 1$.

2. **Unit determinant**: $\det R(\mathbf{q}) = +1$, confirming that $R(\mathbf{q}) \in \mathrm{SO}(3)$.

Note that $\mathbf{q}$ and $-\mathbf{q}$ yield the same rotation matrix $R(\mathbf{q}) = R(-\mathbf{q})$, reflecting the *double cover* of SO(3) by the unit quaternions.

Table 2: Comparison of rotation representations.

Property	Euler	Matrix	Quaternion
Parameters	3	9	4
Gimbal lock	Yes	No	No
Singularity-free	No	Yes	Yes
Composition cost	High	$O(27)$	$O(16)$
Renormalisation	—	$O(n^3)$	$O(n)$
Interpolation (SLERP)	Poor	Poor	Native

3.6. Quaternion Integration from Angular Velocity

Given the current orientation quaternion $\mathbf{q}(t)$ and a gyroscope reading $\boldsymbol{\omega}_b = (\omega_x, \omega_y, \omega_z)$, the orientation evolves according to the differential equation:

$$\dot{\mathbf{q}}(t) = \frac{1}{2}\,\mathbf{q}(t) \otimes \boldsymbol{\omega}_q \tag{43}$$

where $\boldsymbol{\omega}_q = (0, \omega_x, \omega_y, \omega_z)$ is the angular velocity expressed as a pure quaternion. Using a first-order discretisation with time step Δt:

$$\mathbf{q}(t + \Delta t) = \mathbf{q}(t) + \frac{\Delta t}{2}\,\mathbf{q}(t) \otimes \boldsymbol{\omega}_q \tag{44}$$

After each update, $\mathbf{q}$ must be renormalised to remain on the unit sphere: $\mathbf{q} \leftarrow \mathbf{q}/\|\mathbf{q}\|$.

3.7. Advantages over Euler Angles and Rotation Matrices

Table 2 summarises the key properties of the three most common rotation representations. Quaternions offer the best trade-off between compactness, numerical stability, and computational cost for real-time sensor fusion applications.

Euler angles, while intuitive, suffer from *gimbal lock*—a singularity that occurs when two rotation axes become aligned, causing a loss of one degree of freedom. Rotation matrices are singularity-free but use nine parameters (with six orthogonality constraints), making renormalisation expensive. Quaternions use only four parameters (with one constraint $\|\mathbf{q}\| = 1$), avoid gimbal lock entirely, compose efficiently via the Hamilton product, and support smooth interpolation through spherical linear interpolation (SLERP). These properties make quaternions the natural choice for the sensor fusion pipeline developed in this work.

4. Complementary Filter for Attitude Estimation

4.1. Intuition

As discussed in Section 2.5, the accelerometer and gyroscope have complementary noise characteristics. The key insight is that their useful information occupies *different frequency bands*:

- The accelerometer-derived angle is reliable at **low frequencies** (below ∼1 Hz) but corrupted at high frequencies.
- The gyroscope-integrated angle is reliable at **high frequencies** but drifts at low frequencies.

A complementary filter combines the two by applying a *low-pass filter* to the accelerometer and a *high-pass filter* to the gyroscope, such that their transfer functions sum to unity across all frequencies.

4.2. Filter Derivation

Let $\theta_a(t)$ denote the tilt angle estimated from the accelerometer and $\theta_g(t)$ the angle obtained by integrating the gyroscope. The complementary filter output is:

$$\boxed{\hat{\theta}(t) = \alpha\big[\hat{\theta}(t - \Delta t) + \omega(t)\,\Delta t\big] + (1 - \alpha)\,\theta_a(t)} \tag{45}$$

where $\alpha \in (0, 1)$ is the filter coefficient. The first term is a *high-pass filter* on the gyroscope (it propagates the previous estimate forward and adds the incremental rotation), and the second term is a *low-pass filter* on the accelerometer (it slowly corrects toward the gravity reference).

The time constant of the filter is:

$$\tau = \frac{\alpha\,\Delta t}{1 - \alpha} \tag{46}$$

Typical values are $\alpha = 0.96$–0.98, corresponding to time constants of a few seconds.

4.3. Accelerometer Tilt Extraction

When the sensor is approximately stationary, the accelerometer measures the gravity vector projected onto the body axes. The roll and pitch angles can be extracted as:

$$\phi = \arctan\left(\frac{a_y}{\sqrt{a_x^2 + a_z^2}}\right) \tag{47}$$

$$\theta = \arctan\left(\frac{-a_x}{\sqrt{a_y^2 + a_z^2}}\right) \tag{48}$$

These relations assume that the only force acting on the sensor is gravity. During dynamic motion, the estimates become unreliable—precisely the regime where the gyroscope takes over through the high-pass path.

4.4. Quaternion Form of the Complementary Filter

To avoid gimbal lock, the complementary filter is implemented in quaternion space. At each time step:

1. Integrate the gyroscope to obtain a predicted quaternion $\mathbf{q}_{\text{pred}}$ using Eq. (44).
2. Compute a reference quaternion $\mathbf{q}_{\text{acc}}$ from the accelerometer tilt angles (Eqs. 47–48).
3. Blend the two using spherical linear interpolation (SLERP):

$$\mathbf{q}(t) = \text{SLERP}\big(\mathbf{q}_{\text{acc}},\ \mathbf{q}_{\text{pred}},\ \alpha\big) \tag{49}$$

This is the quaternion analogue of the scalar Eq. (45), with the SLERP parameter playing the role of α.

5. Gradient-Descent Attitude Estimation: The Madgwick Filter

The complementary filter presented in Section 4 provides an effective and intuitive approach to sensor fusion. However, its scalar formulation operates on Euler angles independently, which introduces two practical limitations: (i) the relationship between body-frame angular velocities and Euler angle rates is *nonlinear and coupled* (requiring explicit Euler rate equations), and (ii) the representation becomes singular near $\pm 90°$ pitch (gimbal lock). While the quaternion form of the complementary filter (Section 4.4) addresses these issues in principle via SLERP blending, an alternative family of algorithms performs the fusion *natively* in quaternion space using gradient descent on an orientation error objective. The most widely adopted of these is the filter proposed by Madgwick et al. (2011), which we describe in this section.

5.1. Problem Statement

Given a stream of gyroscope readings $\boldsymbol{\omega}_b \in \mathbb{R}^3$, accelerometer readings $\mathbf{a}_b \in \mathbb{R}^3$, and (optionally) magnetometer readings $\mathbf{m}_b \in \mathbb{R}^3$, we seek the unit quaternion $\mathbf{q}(t)$ that represents the orientation of the sensor body frame with respect to a fixed world frame at each time step.

The key idea is to formulate attitude estimation as an *optimisation problem*: find the quaternion that best aligns the sensor measurements with known reference directions in the world frame (gravity and the geomagnetic field).

5.2. Gyroscope Prediction

As in the complementary filter, the gyroscope provides a short-term orientation prediction. Given the current estimate $\mathbf{q}(t)$ and the angular velocity $\boldsymbol{\omega}_b(t)$, the predicted quaternion is obtained by integrating the quaternion kinematic equation (Eq. 44):

$$\mathbf{q}_\omega(t + \Delta t) = \mathbf{q}(t) + \frac{\Delta t}{2}\,\mathbf{q}(t) \otimes \boldsymbol{\omega}_q(t) \tag{50}$$

where $\boldsymbol{\omega}_q = (0,\, \omega_x,\, \omega_y,\, \omega_z)$. This step is identical to the gyroscope path in the complementary filter and inherits the same drift characteristics.

5.3. Accelerometer and Magnetometer Correction via Gradient Descent

The correction step is where the Madgwick filter fundamentally differs from the complementary filter. Instead of extracting Euler angles from the accelerometer and blending them with the gyroscope output, the Madgwick filter defines an *objective function* whose minimum corresponds to the correct orientation.

5.3.1 Objective Function (IMU: Accelerometer Only)

Let $\hat{\mathbf{d}}_w = (0,\, 0,\, 1)^\top$ be the expected direction of gravity in the world frame (normalised). If the true orientation is $\mathbf{q}$, then the accelerometer should measure:

$$\hat{\mathbf{a}}_b = \mathbf{q}^* \otimes \hat{\mathbf{d}}_{w,q} \otimes \mathbf{q} \tag{51}$$

The error between the predicted and actual normalised accelerometer reading $\hat{\mathbf{a}}_b^{\text{meas}}$ defines the objective function:

$$f(\mathbf{q}) = \mathbf{q}^* \otimes \hat{\mathbf{d}}_{w,q} \otimes \mathbf{q} - \hat{\mathbf{a}}_{b,q}^{\text{meas}} \tag{52}$$

where $f : \mathbb{R}^4 \to \mathbb{R}^3$ maps a quaternion to the residual between the expected and observed gravity direction. The optimal orientation satisfies $f(\mathbf{q}) = \mathbf{0}$.

5.3.2 Gradient Step

Rather than solving the full nonlinear optimisation, Madgwick applies a *single gradient descent step* per time sample, which is sufficient because the orientation changes slowly relative to the sampling rate:

$$\mathbf{q}_\nabla(t) = \mathbf{q}(t) - \beta\,\frac{\nabla f(\mathbf{q})}{\|\nabla f(\mathbf{q})\|} \tag{53}$$

where $\nabla f = J^\top(\mathbf{q})\, f(\mathbf{q})$ is computed analytically from the Jacobian $J(\mathbf{q})$ of the objective function with respect to the quaternion components. The normalisation $\|\nabla f\|$ ensures that the step magnitude depends only on β, not on the error amplitude.

5.3.3 Extension to MARG (with Magnetometer)

When magnetometer data is available, the objective function is extended to include the geomagnetic field direction. Let $\hat{\mathbf{b}}_w$ be the expected magnetic field direction in the world frame (computed from the magnetometer measurement itself, projected to have zero East component). The combined objective becomes:

$$f_{\text{MARG}}(\mathbf{q}) = \begin{bmatrix} f_g(\mathbf{q}) \\ f_b(\mathbf{q}) \end{bmatrix} \tag{54}$$

Table 3: Complementary filter vs. Madgwick filter.

Property	Complementary	Madgwick
Fusion domain	Euler angles	Quaternion
Gimbal lock	Yes (scalar form)	No
Euler rate equations	Required	Not needed
Magnetometer fusion	Separate filter	Integrated
Tuning parameters	α per axis	Single β
Computational cost	Very low	Low
Gravity sign convention	Must be handled	Automatic

where f_g is the gravity residual and f_b is the magnetic field residual. The Jacobian becomes a 6×4 matrix, and the gradient step proceeds analogously. The magnetometer provides an absolute heading reference that the accelerometer alone cannot observe (since gravity is invariant under rotations about the vertical axis, as discussed in Section 6.2).

5.4. Filter Fusion

The final quaternion estimate is a weighted combination of the gyroscope prediction and the gradient correction:

$$\boxed{\mathbf{q}(t+\Delta t) = \gamma\, \mathbf{q}_\omega(t+\Delta t) + (1-\gamma)\, \mathbf{q}_\nabla(t+\Delta t)} \tag{55}$$

followed by renormalisation $\mathbf{q} \leftarrow \mathbf{q}/\|\mathbf{q}\|$. In the original formulation, the gain parameter β simultaneously controls the step size and the effective blending ratio, playing a role analogous to α in the complementary filter. A typical value is $\beta = 0.041$ for MARG configurations.

The crucial difference is that **the entire fusion operates in quaternion space**: there is no intermediate conversion to Euler angles, no need for Euler rate equations, and no singularity at $\pm 90°$ pitch.

5.5. Comparison with the Complementary Filter

Table 3 summarises the key differences between the two approaches.

5.5.1 Euler Rate Coupling

In the scalar complementary filter, the gyroscope rates $(\omega_x, \omega_y, \omega_z)$ are used directly as the rates of change of roll, pitch, and yaw. This is only valid for small angles. The correct relationships are:

$$\dot{\phi} = \omega_x + \sin\phi\, \tan\theta\, \omega_y + \cos\phi\, \tan\theta\, \omega_z \tag{56}$$

$$\dot{\theta} = \cos\phi\, \omega_y - \sin\phi\, \omega_z \tag{57}$$

$$\dot{\psi} = \frac{\sin\phi}{\cos\theta}\, \omega_y + \frac{\cos\phi}{\cos\theta}\, \omega_z \tag{58}$$

At large inclinations (e.g. $\theta > 20°$), ignoring this coupling introduces significant cross-axis errors that corrupt the orientation estimate and, consequently, the gravity removal step. The Madgwick filter avoids this entirely because quaternion integration (Eq. 50) naturally handles the nonlinear coupling between axes without any explicit trigonometric corrections.

5.5.2 Gravity Sign Convention

A subtle but practically important difference concerns the sign of the gravity vector. The complementary filter extracts tilt angles from the accelerometer and subtracts gravity as a scalar magnitude $\mathbf{g}_w = (0, 0, \pm g)$, where the sign depends on the sensor convention. If the sensor reports $a_z \approx +9.8$ m/s^2 at rest (screen up), the gravity vector must be subtracted as $(0, 0, +g)$; if it reports $a_z \approx -9.8$ m/s^2 (screen down or alternate convention), the sign must be flipped.

The Madgwick filter is immune to this issue: it works with the *normalised* accelerometer reading and optimises for the direction of gravity, not its sign. The gravity vector in the world frame is then determined empirically from the calibration phase as:

$$\mathbf{g}_w = \frac{1}{N_c} \sum_{n=1}^{N_c} \mathbf{a}_w[n] \tag{59}$$

where $\mathbf{a}_w[n] = \mathbf{q}[n] \otimes \mathbf{a}_{b,q}[n] \otimes \mathbf{q}^*[n]$ is the accelerometer reading rotated to the world frame using the Madgwick quaternion. This approach is robust to any sensor sign convention and eliminates a common source of implementation errors.

5.5.3 Effect on Position Estimation

Because the Madgwick filter produces a more accurate orientation estimate (especially at large inclinations), the gravity removal step is more precise, which directly reduces the gravity leakage that dominates the position drift. In our experiments with a 39-second recording of repetitive vertical hand movements ("subidas_bajadas_con_pausa"), the Madgwick pipeline produced:

- A Z-axis range of [0.00, 0.76] m, consistent with the actual hand movement amplitude.
- No negative Z values, confirming that the hand never descended below its starting position (as expected from the experimental protocol).
- Near-zero residual bias after gravity subtraction ($\sim 10^{-6}$ m/s^2), compared to biases of order ± 19 m/s^2 with the scalar complementary filter before sign correction.

Furthermore, the Madgwick pipeline does not require a high-pass filter on the linear acceleration, which was previously necessary to compensate for gravity

leakage in the complementary filter approach. The removal of this filter is significant because the high-pass filter, while effective at removing drift, also *destroys sustained displacements*: it forces the acceleration to have zero mean, which means any net position change (e.g. moving up and staying up) is gradually pulled back toward the origin. This manifested as backward loops at rest positions and spurious negative heights—artefacts that disappear entirely with the improved orientation from the Madgwick filter.

6. World-Frame Projection and Position Integration

6.1. Rotating Acceleration to the World Frame

Once the attitude quaternion $\mathbf{q}(t)$ has been estimated by the complementary filter, each raw accelerometer reading $\mathbf{a}_b(t)$ can be transformed into the world frame:

$$\mathbf{a}_w(t) = \mathbf{q}(t) \otimes \mathbf{a}_{b,q}(t) \otimes \mathbf{q}^*(t) \tag{60}$$

where $\mathbf{a}_{b,q} = (0, a_x, a_y, a_z)$ is the accelerometer reading embedded as a pure quaternion. The vector part of the result gives $\mathbf{a}_w = (a_{w,x},\ a_{w,y},\ a_{w,z})$ in the world frame.

6.2. Gravity Removal

In the world frame, gravity appears as a constant vector $\mathbf{g}_w = (0,\ 0,\ g)^\top$ (assuming ENU convention with $g \approx 9.81$ m/s^2). The linear acceleration is:

$$\mathbf{a}_{\text{lin}}(t) = \mathbf{a}_w(t) - \mathbf{g}_w \tag{61}$$

This step is possible *only* because the rotation to the world frame has been performed first. In the body frame, the gravity component changes direction with every rotation, making subtraction impossible.

6.3. Double Integration to Position

Velocity and position are obtained by numerical integration of the gravity-free acceleration:

$$\mathbf{v}(t + \Delta t) = \mathbf{v}(t) + \mathbf{a}_{\text{lin}}(t)\,\Delta t \tag{62}$$

$$\mathbf{p}(t + \Delta t) = \mathbf{p}(t) + \mathbf{v}(t)\,\Delta t + \tfrac{1}{2}\,\mathbf{a}_{\text{lin}}(t)\,\Delta t^2 \tag{63}$$

with initial conditions $\mathbf{v}(0) = \mathbf{0}$ and $\mathbf{p}(0) = \mathbf{0}$ (the sensor starts at rest at the origin).

6.4. Drift in Position Estimation

Double integration amplifies any residual error in $\mathbf{a}_{\text{lin}}$: a constant bias ϵ in acceleration produces a position error that grows as $\frac{1}{2}\epsilon\, t^2$. This error has two distinct sources:

Static bias. The accelerometer has a small, approximately constant offset that persists even at rest. This can be estimated and removed during an initial calibration phase by averaging the linear acceleration over a stationary period of several seconds.

Gravity leakage from orientation error. This is the dominant source of drift. Even a small error δ in the estimated roll or pitch causes the gravity vector to be imperfectly subtracted after the world-frame rotation, leaving a residual:

$$\epsilon_g \approx g \sin(\delta) \tag{64}$$

For example, an orientation error of just $\delta = 1.4$ř produces a residual of $9.81 \times \sin(1.4$ř$) \approx 0.24$ m/s^2. After double integration over a time interval T:

$$\Delta p = \tfrac{1}{2}\,\epsilon_g\, T^2 = \tfrac{1}{2}\, g \sin(\delta)\, T^2 \tag{65}$$

For $T = 3$ s, this yields $\Delta p \approx 1.08$ m of spurious displacement. Crucially, this error *changes direction* as the device rotates—it is not a fixed offset that can be calibrated away. It affects all three world-frame axes (X, Y, Z), not just the vertical. This is the fundamental limitation of inertial navigation with consumer-grade MEMS sensors.

6.5. Drift Mitigation Strategies

Three complementary strategies are applied to reduce the drift:

6.5.1 Static Calibration with Measured Gravity

Rather than using the textbook value $g = 9.81$ m/s^2, the actual gravity magnitude is estimated from the sensor data during the initial rest phase:

$$g_{\text{meas}} = \frac{1}{N_c} \sum_{n=1}^{N_c} \|\mathbf{a}_b[n]\| \tag{66}$$

where N_c is the number of samples in the rest phase. The residual linear acceleration bias is then estimated as the mean of $\mathbf{a}_{\text{lin}}$ over the same interval and subtracted from the entire signal. Using the full rest phase ($N_c \approx 1000$ samples at 100 Hz, corresponding to ∼10 s) instead of a handful of samples dramatically improves the bias estimate.

6.5.2 Zero-Velocity Updates (ZUPT)

When the sensor is stationary, its true velocity is zero. Any non-zero integrated velocity at that instant is, by definition, accumulated error. A *zero-velocity update* exploits this constraint by resetting the velocity to zero whenever the sensor is detected to be at rest.

Stationary detection. Two conditions must hold simultaneously over a sliding window of W consecutive samples:

$$\big|\, \|\mathbf{a}_b[n]\| - g \,\big| < \sigma_a \tag{67}$$

$$\|\boldsymbol{\omega}_b[n]\| < \sigma_\omega \tag{68}$$

The first condition checks that the accelerometer norm is close to gravity (no linear acceleration). The second checks that the angular velocity is near zero (no rotation). Requiring both conditions over a window of W samples (typically $W = 5$) prevents false positives from momentary pauses in a single channel. Typical thresholds are $\sigma_a = 0.3$ m/s^2 and $\sigma_\omega = 0.15$ rad/s.

Velocity reset. At every sample classified as stationary, the integrated velocity is forced to zero on all three axes:

$$\text{if stationary}[n]: \quad \mathbf{v}[n] = \mathbf{0} \tag{69}$$

This prevents unbounded velocity drift during rest periods and provides anchor points for the subsequent backward correction.

6.5.3 Backward Velocity Correction

ZUPT alone resets velocity at rest, but during movement segments the velocity still drifts due to gravity leakage. The *backward correction* addresses this by distributing the accumulated velocity error linearly across each movement segment.

Segment identification. A movement segment is defined as the interval between two consecutive stationary phases. Let n_s be the first moving sample (where the sensor leaves rest) and n_e be the last moving sample (just before the next rest phase begins). At n_e, the velocity $\mathbf{v}[n_e]$ should ideally transition smoothly to zero, but in practice it carries an error $\mathbf{e}_v = \mathbf{v}[n_e]$ due to drift accumulated during the segment.

Linear error distribution. The error is assumed to grow approximately linearly during the segment (since gravity leakage acts as a roughly constant bias over short intervals). It is therefore distributed proportionally:

$$\mathbf{v}_{\text{corr}}[n] = \mathbf{v}[n] - \mathbf{e}_v \cdot \frac{n - n_s + 1}{n_e - n_s + 1}, \quad n \in [n_s,\, n_e] \tag{70}$$

At $n = n_s$ the correction is minimal; at $n = n_e$ the full error is subtracted, ensuring a smooth transition to zero velocity at the next ZUPT. The corrected velocity is then integrated to obtain the position estimate.

Limitations. The backward correction is effective when the user's motion naturally includes pauses (e.g. drawing shapes with intermittent stops). However, during continuous uninterrupted motion, no ZUPT events occur and the correction cannot be applied. This motivates the high-pass filtering strategy described next.

6.5.4 High-Pass Filtering of Linear Acceleration

Even after gravity subtraction and static bias removal, the linear acceleration retains a slowly varying residual caused by orientation estimation errors. As shown in Section 6.4, an orientation error of δ produces a gravity leakage $\epsilon_g \approx g \sin(\delta)$ that changes direction as the device rotates but maintains a *non-zero mean* over time. This non-zero mean, when integrated twice, produces the characteristic monotonic drift observed in position estimates.

The key insight is that **real translational acceleration has zero mean**: every acceleration phase (push) is followed by a deceleration phase (stop), so the integral over a complete movement cycle is approximately zero. In contrast, gravity leakage is a low-frequency bias with non-zero mean. A high-pass filter can therefore separate the two.

First-order IIR high-pass filter. We apply a causal first-order high-pass filter independently to each axis of $\mathbf{a}_{\text{lin}}$:

$$y[n] = \alpha_{\text{hp}} \big(y[n-1] + x[n] - x[n-1] \big) \tag{71}$$

where $x[n]$ is the input (linear acceleration after bias removal) and $y[n]$ is the filtered output. The parameter $\alpha_{\text{hp}} \in (0,\, 1)$ controls the cutoff frequency:

$$f_c = \frac{1 - \alpha_{\text{hp}}}{2\pi\, \Delta t} \tag{72}$$

Signals below f_c are attenuated (including the gravity leakage bias); signals above f_c pass through (the actual motion).

Choosing the cutoff frequency. The cutoff must satisfy two constraints simultaneously:

1. f_c must be **above** the frequency of the gravity leakage bias, which is quasi-static ($\ll 1$ Hz).
2. f_c must be **below** the fundamental frequency of the actual motion, so that the real acceleration signal is preserved.

The optimal value therefore depends on the *type of motion*, not on the individual recording:

- **Fast hand gestures** ($\sim$2–4 Hz): $\alpha_{\text{hp}} \approx 0.85$, giving $f_c \approx 2.4$ Hz at 100 Hz sampling.

- **Walking or slow movements** (∼1–2 Hz): $\alpha_{\text{hp}} \approx 0.95$, giving $f_c \approx 0.8$ Hz.
- **Very slow gestures** (< 1 Hz): $\alpha_{\text{hp}} \approx 0.98$, giving $f_c \approx 0.3$ Hz. At this point the filter provides minimal benefit, as the leakage frequency overlaps with the motion frequency.

If f_c is set too high, the filter removes part of the real acceleration signal, causing the estimated displacement to be *underestimated*. If f_c is set too low, the gravity leakage passes through and drift reappears. In practice, α_{hp} does not need to be re-tuned for each recording—it only needs to match the general category of motion (fast vs. slow).

Effect on drift. In our experiments, adding the high-pass filter reduced the position drift from several meters to the order of ±0.2 m for a 17-second recording of repetitive up-and-down hand motion, even without any ZUPT corrections (no pauses in the motion). The mean residual acceleration on the Z-axis dropped from +0.41 m/s^2 (without filter) to +0.02 m/s^2 (with $\alpha_{\text{hp}} = 0.85$).

6.5.5 Summary of the Drift Mitigation Pipeline

The complete pipeline applies five corrections in sequence:

1. **Measured gravity magnitude**: g_{meas} from the rest phase replaces the nominal 9.81 m/s^2.
2. **Static bias removal**: mean of $\mathbf{a}_{\text{lin}}$ during rest is subtracted.
3. **Moving average**: a short window (∼5 samples) smooths high-frequency noise and vibration spikes.
4. **High-pass filter**: removes the low-frequency gravity leakage bias that causes monotonic drift.
5. **Acceleration clamp**: limits $|\mathbf{a}_{\text{lin}}|$ to a physically reasonable maximum (e.g. ±4 m/s^2 for hand motion), rejecting residual spikes from large orientation errors.

After filtering, ZUPT and backward velocity correction are applied during integration when stationary phases are available. For continuous motion without pauses, the high-pass filter alone provides the primary drift mitigation.

7. Experimental Results

8. Conclusions and Future Work

A. Derivation of Gyroscope Integration Error Bounds

This appendix provides a rigorous derivation of the three error terms that arise when numerically integrating a gyroscope signal: bias drift, angular random walk, and discretisation error. These results underpin the theoretical envelopes shown in Figs. 2 and 3.

A.1. Setup and Notation

Consider a single-axis gyroscope sampled at frequency $f_s = 1/\Delta t$. At each time step n, the measured angular velocity is:

$$\omega[n] = \omega_{\text{true}}[n] + b_\omega + n_\omega[n] \tag{73}$$

where:

- $\omega_{\text{true}}[n]$ is the true angular velocity,
- b_ω is a constant bias (deg/s),
- $n_\omega[n]$ are independent and identically distributed (i.i.d.) random variables with $E[n_\omega[n]] = 0$ and $\text{Var}[n_\omega[n]] = \sigma_\omega^2$.

The angle is estimated by rectangular (Euler) integration:

$$\hat{\theta}[N] = \sum_{n=1}^{N} \omega[n]\,\Delta t \tag{74}$$

while the true angle is:

$$\theta_{\text{true}}[N] = \sum_{n=1}^{N} \omega_{\text{true}}[n]\,\Delta t \tag{75}$$

The total error is the difference:

$$\epsilon[N] = \hat{\theta}[N] - \theta_{\text{true}}[N] = \sum_{n=1}^{N} \big(b_\omega + n_\omega[n]\big)\,\Delta t \tag{76}$$

We can separate this into two independent terms:

$$\epsilon[N] = \underbrace{\sum_{n=1}^{N} b_\omega\,\Delta t}_{\epsilon_{\text{bias}}[N]} + \underbrace{\sum_{n=1}^{N} n_\omega[n]\,\Delta t}_{\epsilon_{\text{rw}}[N]} \tag{77}$$

A.2. Bias Drift: ϵ_{bias}

Since b_ω is constant, the sum is trivial:

$$\epsilon_{\text{bias}}[N] = \sum_{n=1}^{N} b_\omega\,\Delta t = N \cdot b_\omega \cdot \Delta t \tag{78}$$

Substituting $T = N\,\Delta t$ (total elapsed time):

$$\boxed{\epsilon_{\text{bias}} = b_\omega \cdot T} \tag{79}$$

Properties.

- **Deterministic**: the bias drift is the same in every realisation of the experiment.
- **Linear in time**: doubling the measurement duration doubles the error.
- **Independent of Δt**: for a fixed total time T, changing the sampling frequency does not affect the bias drift. This is because $N \cdot \Delta t = T$ regardless of how we partition the interval.

A.3. Angular Random Walk: ϵ_{rw}

This is the stochastic term and requires a careful probabilistic analysis.

A.3.1 Step 1: Write the sum explicitly

$$\epsilon_{\text{rw}}[N] = \sum_{n=1}^{N} n_\omega[n]\,\Delta t = \Delta t \sum_{n=1}^{N} n_\omega[n] \tag{80}$$

A.3.2 Step 2: Compute the expected value (mean)

Since $E[n_\omega[n]] = 0$ for all n, and expectation is linear:

$$\begin{aligned} E\big[\epsilon_{\text{rw}}[N]\big] &= \Delta t \sum_{n=1}^{N} E[n_\omega[n]] \\ &= \Delta t \sum_{n=1}^{N} 0 \\ &= 0 \end{aligned} \tag{81}$$

The random walk has zero mean—it does not favour any direction on average.

A.3.3 Step 3: Compute the variance

The variance of a sum of *independent* random variables equals the sum of their variances. This is the key property we exploit. Since $n_\omega[n]$ are i.i.d.:

$$\begin{aligned} \operatorname{Var}\big[\epsilon_{\text{rw}}[N]\big] &= \operatorname{Var}\left[\Delta t \sum_{n=1}^{N} n_\omega[n]\right] \\ &= \Delta t^2 \operatorname{Var}\left[\sum_{n=1}^{N} n_\omega[n]\right] \end{aligned} \tag{82}$$

where we used the property $\operatorname{Var}[cX] = c^2 \operatorname{Var}[X]$ to pull out the constant Δt^2. Now, for independent variables:

$$\begin{aligned} \operatorname{Var}\left[\sum_{n=1}^{N} n_\omega[n]\right] &= \sum_{n=1}^{N} \operatorname{Var}[n_\omega[n]] \\ &= \sum_{n=1}^{N} \sigma_\omega^2 \\ &= N\,\sigma_\omega^2 \end{aligned} \tag{83}$$

Substituting into Eq. (82):

$$\operatorname{Var}\big[\epsilon_{\text{rw}}[N]\big] = \Delta t^2 \cdot N \cdot \sigma_\omega^2 = N\,\Delta t^2\,\sigma_\omega^2 \tag{84}$$

A.3.4 Step 4: Compute the standard deviation

The standard deviation (the "1-sigma envelope") is the square root of the variance:

$$\sigma_{\epsilon_{\text{rw}}} = \sqrt{N\,\Delta t^2\,\sigma_\omega^2} = \sigma_\omega\,\Delta t\,\sqrt{N} \tag{85}$$

A.3.5 Step 5: Express in terms of physical quantities

Substituting $N = T/\Delta t$:

$$\begin{aligned} \sigma_{\epsilon_{\text{rw}}} &= \sigma_\omega\,\Delta t\,\sqrt{\frac{T}{\Delta t}} \\ &= \sigma_\omega\,\Delta t \cdot \frac{\sqrt{T}}{\sqrt{\Delta t}} \\ &= \sigma_\omega\,\sqrt{\Delta t}\,\sqrt{T} \\ &= \sigma_\omega\,\sqrt{\Delta t \cdot T} \end{aligned} \tag{86}$$

$$\boxed{\sigma_{\epsilon_{\text{rw}}} = \sigma_\omega\,\sqrt{\Delta t \cdot T}} \tag{87}$$

Interpretation.

- The standard deviation grows as $\sqrt{T}$—much slower than the linear bias drift. This is the hallmark of a *random walk*.
- Unlike the bias drift, the random walk *does* depend on Δt: a higher sampling frequency (smaller Δt) yields a *smaller* random walk, because each noise sample is multiplied by a smaller time step before being accumulated.
- The $\pm 1\sigma$ envelope contains approximately 68% of the realisations (assuming Gaussian noise). The $\pm 2\sigma$ envelope contains ~95%.

A.3.6 Relation to the Angular Random Walk coefficient

In sensor datasheets, manufacturers report the *Angular Random Walk* (ARW) coefficient, defined as:

$$\sigma_{\text{ARW}} = \sigma_\omega \sqrt{\Delta t} = \frac{\sigma_\omega}{\sqrt{f_s}} \qquad (\text{units: deg}/\sqrt{\text{s}}) \tag{88}$$

so that Eq. (87) simplifies to:

$$\sigma_{\epsilon_{\text{rw}}} = \sigma_{\text{ARW}} \sqrt{T} \tag{89}$$

This compact form is commonly used in inertial navigation literature.

A.4. Discretisation Error

Even if the gyroscope were *perfect* ($b_\omega = 0$, $\sigma_\omega = 0$), a third error arises from approximating a continuous integral with a discrete sum. We derive its order using Taylor expansion.

A.4.1 Step 1: Local truncation error

The true integral over one time step is:

$$\int_{t_n}^{t_n+\Delta t} \omega(\tau)\, d\tau \tag{90}$$

The rectangular (Euler) rule approximates this as $\omega(t_n)\,\Delta t$. Expanding $\omega(\tau)$ in a Taylor series around t_n:

$$\omega(\tau) = \omega(t_n) + \dot{\omega}(t_n)(\tau - t_n) + \mathcal{O}\big((\tau - t_n)^2\big) \tag{91}$$

Integrating from t_n to $t_n + \Delta t$:

$$\begin{aligned}\int_{t_n}^{t_n+\Delta t} \omega(\tau)\, d\tau &= \omega(t_n)\,\Delta t + \dot{\omega}(t_n)\,\frac{\Delta t^2}{2} + \mathcal{O}(\Delta t^3) \\ &= \underbrace{\omega(t_n)\,\Delta t}_{\text{Euler approx.}} + \underbrace{\dot{\omega}(t_n)\,\frac{\Delta t^2}{2} + \mathcal{O}(\Delta t^3)}_{\text{local truncation error}}\end{aligned} \tag{92}$$

Therefore, the local truncation error per step is:

$$e_{\text{local}} = \dot{\omega}(t_n)\,\frac{\Delta t^2}{2} = \mathcal{O}(\Delta t^2) \tag{93}$$

A.4.2 Step 2: Global accumulated error

Over $N = T/\Delta t$ steps, the errors accumulate:

$$\begin{aligned}\epsilon_{\text{disc}} &= \sum_{n=1}^{N} e_{\text{local}}[n] \\ &= \sum_{n=1}^{N} \mathcal{O}(\Delta t^2) \\ &= N \cdot \mathcal{O}(\Delta t^2) \\ &= \frac{T}{\Delta t} \cdot \mathcal{O}(\Delta t^2)\end{aligned} \tag{94}$$

$$\boxed{\epsilon_{\text{disc}} = \mathcal{O}(\Delta t)} \tag{95}$$

Interpretation.

- The global discretisation error is **first-order** in Δt. Halving the time step (doubling the sampling frequency) halves this error.
- For a sensor sampling at $f_s = 100$ Hz ($\Delta t = 0.01$ s), this error is typically very small compared to the bias drift, which is why increasing the sampling rate has a marginal effect in practice.

A.5. Complete Error Budget

Combining all three terms, the total angle error after time T is:

$$\epsilon_{\text{total}}(T) = \underbrace{b_\omega \cdot T}_{\text{bias drift}} + \underbrace{\sigma_\omega \sqrt{\Delta t \cdot T}}_{\text{random walk (stochastic)}} + \underbrace{\mathcal{O}(\Delta t)}_{\text{discretisation}} \tag{96}$$

Table 4 compares the scaling of each term with respect to both elapsed time T and sampling interval Δt.

Table 4: Scaling of the three gyroscope integration error terms.

Error term	Grows with T	Depends on Δt?	Domi
Bias drift	$\propto T$ (linear)	No	Yes (lon
Random walk	$\propto \sqrt{T}$	Yes ($\propto \sqrt{\Delta t}$)	Med
Discretisation	Fixed for fixed T	Yes ($\propto \Delta t$)	Negli

Numerical example. For a typical MEMS gyroscope with $b_\omega = 0.05$ deg/s, $\sigma_\omega = 0.1$ deg/s, at $f_s = 100$ Hz ($\Delta t = 0.01$ s):

	$T = 1$ s	$T = 10$ s	$T = 60$ s	$T = 300$ s
ϵ_{bias}	0.05ř	0.50ř	3.00ř	15.00ř
$\sigma_{\epsilon_{\text{rw}}}$	0.01ř	0.03ř	0.08ř	0.17ř
Ratio (bias/rw)	5×	16×	39×	87×

The bias drift dominates increasingly with time: after 5 minutes the bias error is 87 times larger than the random walk. This justifies the use of external corrections (complementary filter, ZUPT) to bound the drift.